

Disproof of Conjecture 00000000407

The stated equal-size Littlewood–Richardson table is zero

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Abstract

Both language versions index all three partitions by the same integer n . For this exact indexing, every Littlewood–Richardson coefficient is zero when $n > 0$. We define and count conventional Littlewood–Richardson tableaux, derive their size identity, and refute the complete positive-constant asymptotic assertion. The accompanying Lean project proves these statements for every n , including finiteness of the counted types. Independent exact computations provide additional finite checks of the definitions.

1 The exact assertion

The official English statement is:

Conjecture: The number of nonzero entries of the LR coefficient table $(c_{\lambda\mu}^\nu)_{\lambda,\mu,\nu \vdash n}$ is asymptotically $c \cdot 4^n/n^{5/4}$ (a density estimate for LR-realizable triples among all 4^n configurations).

The Chinese text has the same simultaneous indexing and quantitative formula. The full original bilingual file accompanies this submission as `conjecture.md`. Its source identities are:

Upstream commit	ba46ce0997cbf232fe95e5dced37ff4d32e1eb94
Source Git blob	3439f91b71e05a0d57573e84e6c181d4c7a0ba86

Keep the indexing as printed, and write

$$A(n) = \#\{(\lambda, \mu, \nu) \in \text{Part}(n)^3 : c_{\lambda\mu}^\nu \neq 0\}, \quad g(n) = \frac{4^n}{n^{5/4}} \quad (n > 0).$$

The triples are ordered and each nonzero entry contributes one, regardless of its coefficient value. The conventional asymptotic claim asserts that some fixed real $c > 0$ satisfies $A(n)/(cg(n)) \rightarrow 1$ as integers $n \rightarrow \infty$.

The parenthetical does not define an ambient configuration family. We do not invent one: the explicit quantitative assertion is necessary whether the parenthetical is explanatory or adds a further requirement. Refuting that assertion therefore suffices. A source-only meaning assessment initially deferred a proof of the entire statement because of this phrase, then cleared this disproof-only scope. No claim is made about a different intended problem obtained by changing the displayed indexing.

2 Actual partitions and LR tableaux

Represent an integer partition by its finite Young diagram, a lower set in $\mathbb{N}^2$ for the coordinatewise order. Coordinates increase downward and to the right. Diagram cardinality is partition size. Mathlib's `YoungDiagram` has exactly this meaning; its row-length equivalence identifies it with decreasing lists of positive parts. Thus the type `Partition n` consists of all diagrams of cardinality n .

For diagrams $\lambda \subseteq \nu$, put $S = \nu \setminus \lambda$. An LR tableau of shape ν/λ and content μ is a filling of S with positive integers satisfying the following conditions:

1. Rows increase weakly from left to right; columns increase strictly downward.
2. Letter $i + 1$ occurs exactly μ_{i+1} times, for every $i \in \mathbb{N}$, including all zero parts beyond the support of μ .
3. Read rows from top to bottom, and each row from right to left. In every prefix, each positive letter occurs at least as often as its successor.

The coefficient $c''_{\lambda\mu}$ is the number of these tableaux; if containment fails, that number is zero. This is the standard combinatorial definition in [1, Definition 2.1] and [2, Section 2]. The references identify the conventional objects; no external vanishing theorem is assumed in the Lean proof.

The formal structure `LRTableau` records the filling, containment, the two monotonicity conditions, all content equations, and the prefix condition. Lean labels are shifted down by one: entry i represents positive letter $i + 1$. No size identity is assumed as a field.

Pairwise row and column inequalities match the adjacent-cell definitions: every skew row and column is an interval, and transitivity connects adjacent comparisons. The formal reading relation is

$$(r, s) \preceq (r', s') \iff r < r' \text{ or } (r = r' \text{ and } s' \leq s).$$

Its order properties are proved. The cells preceding an ending cell form its inclusive reading prefix. Every nonempty prefix has a last cell; the empty prefix satisfies the inequalities automatically. Consequently no reading prefixes or labels are omitted.

3 Finiteness and the size obstruction

Every occurring Lean label i has positive content, so $i < \text{height}(\mu)$. This is derived from the full content equations. A tableau therefore injects into the finite function type $S \rightarrow \{0, \dots, \text{height}(\mu) - 1\}$. Its filling determines it; the proof fields do not create extra objects. Thus the formal `lrcoefficient`, defined using `Nat.card`, counts a finite type and does not exploit the value of cardinality on infinite types.

Similarly, each coordinate of a cell in a size- n lower set is less than n : an occupied cell in row i forces at least $i + 1$ cells, and likewise for columns. Every such diagram belongs to the powerset of an n by n grid. This proves finiteness of `Partition n`, its ordered triple product, and the subtype of nonzero entries counted by `nonzeroCount`.

Lemma 1. *Every LR tableau of shape ν/λ and content μ satisfies $|\nu| = |\lambda| + |\mu|$.*

Proof. Counting a diagram by row-coordinate fibers gives $|\mu| = \sum_{i < \text{height}(\mu)} \mu_{i+1}$. Counting the skew cells by their entry values, using the derived label bound and all content equations, gives

$$|\nu \setminus \lambda| = \sum_{i < \text{height}(\mu)} \#\{a \in S : \text{entry}(a) = i\} = \sum_{i < \text{height}(\mu)} \mu_{i+1} = |\mu|.$$

Containment and finite-set subtraction give $|\nu \setminus \lambda| + |\lambda| = |\nu|$. $\square$

Theorem 2. $A(n) = 0$ for every positive integer n . No fixed positive real constant c satisfies $A(n) \sim c4^n/n^{5/4}$.

Proof. If an entry in the stated table were nonzero, there would be an LR tableau with $|\lambda| = |\mu| = |\nu| = n$. The lemma would imply $n = n + n$, impossible for $n > 0$. The nonzero-entry type is empty, proving $A(n) = 0$.

For completeness, the full ratio assertion is

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, \left| \frac{A(n)}{cg(n)} - 1 \right| < \varepsilon.$$

Take $\varepsilon = 1/2$ and, for any proposed N , take $n = N + 1$. Then $A(n) = 0$ and $cg(n) > 0$, making the absolute difference exactly 1. This contradicts $1 < 1/2$. $\square$

In fact, $A(0) = 1$: the empty tableau is the unique entry in the empty-partition table. This finite initial value has no effect on the limit. The actual normalized sequence $A(n)/g(n)$ is eventually zero and converges to zero.

4 Formal statement and correspondence

The real scale uses `Real.rpow` with $5/4$ explicitly in Lean’s real type, so there is no integer-division interpretation of the exponent. Its positivity for $n > 0$ is proved. `RatioAsymptotic c` is exactly the displayed epsilon- N formula. The final theorem is

`Conjecture407.conjecture407_false`: $\neg \exists c \in \mathbb{R}, 0 < c \wedge \text{RatioAsymptotic}(c)$.

The stronger theorem `no_real_ratio_constant` excludes every real c for the formal ratio predicate. Lean totalizes division at zero; that additional case is immaterial to the ordinary positive-leading-constant claim, whose denominator is positive on the tail.

Formal declaration	Mathematical content
<code>LRTableau.entry_lt</code>	Derived letter bound and finite tableau count
<code>lrTableauFinite</code>	
<code>diagram_card_eq_sum_rowLen</code>	Row-by-row diagram cardinality
<code>LRTableau.size_balance</code>	Proved identity $ \nu = \lambda + \mu $
<code>partitionFinite</code>	Finiteness of the exact index and counted-entry types
<code>nonzeroEntryFinite</code>	
<code>lrCoefficient_eq_zero</code>	Every entry and its nonzero count vanish for $n > 0$
<code>nonzeroCount_eq_zero</code>	
<code>normalized_tendsto_zero</code>	Actual normalized limit in Mathlib’s filter formulation
<code>conjecture407_false</code>	Negation of the full positive-constant asymptotic claim

`Audit407.lean` additionally proves the unit coefficient $c_{\emptyset, \emptyset}^{\emptyset} = 1$ and $A(0) = 1$, $A(1) = A(2) = 0$. These are kernel-checked statements. No Schur-function or representation-theory formalization is used or claimed; the Lean development uses the conventional tableau definition directly.

5 Independent finite checks and reproduction

The separate standard-library Python verifier implements actual LR-tableau enumeration with exact content checks, without a degree-mismatch shortcut. It examines all 116 947 ordered triples for $0 \leq n \leq 10$, giving nonzero counts $1, 0, \dots, 0$. Its independently implemented sparse-polynomial calculation uses Jacobi–Trudi [3, (39)] and alternant coefficient extraction [4, Bi-Alternant Corollary]. Six variables suffice for the tested total input degree at most six.

All 4170 tested triples agree between the two methods. Of these, 1110 have compatible degree, with coefficient frequencies 816 zeros, 293 ones, and one two; the other 3060 have coefficient zero. Fourteen named coefficient checks include units, Pieri cases, unequal input sizes and $c_{(2,1),(2,1)}^{(3,2,1)} = 2$. Five full-filling checks separately exercise validity and isolated row, column, reading-word and content failures. These finite experiments check the implementations; the proof for all n is the Lean argument above and does not depend on them.

The project pins Lean 4.19.0 and Mathlib v4.19.0, including all nine exact dependency revisions in `lean/lake-manifest.json`. From the submission directory, reproduce the mathematical checks with:

```
cd lean
lake build
lake env lean -DwarningAsError=true Conjecture407.lean
lake env lean -DwarningAsError=true Audit407.lean
lake env lean -DwarningAsError=true Check.lean
cd ..
python3 auxiliary/verify.py > actual-output.json
cmp actual-output.json auxiliary/expected-output.json
```

The verification directory supplies source identities, commands and outputs, the full declaration/type/axiom inventory, and the independent review record. Inspection tooling is separate from the mathematical modules. No admitted proof, `native_decide`, or custom axiom is used. Local validation and the independent review are submission evidence; maintainer acceptance is a separate decision.

References

- [1] H. Hahn, *On Classical groups detected by the triple tensor product and the Littlewood–Richardson semi-group*, Research in Number Theory 2, article 19 (2016), Definition 2.1. doi:10.1007/s40993-016-0049-3.
- [2] C. Gutschwager, *On Base Partitions and Cover Partitions of Skew Characters*, Electronic Journal of Combinatorics 15 (2008), N30, Section 2. Publisher PDF.
- [3] M. A. A. van Leeuwen, *Schur functions and alternating sums*, Electronic Journal of Combinatorics 11(2) (2006), A5, equation (39). Publisher PDF.
- [4] J. R. Stembridge, *A Concise Proof of the Littlewood–Richardson Rule*, Electronic Journal of Combinatorics 9 (2002), N5, Bi-Alternant Corollary. Publisher PDF.